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SkyFlux AI: Scalable Airspace Intelligence and Anomaly Detection via Cloud-Native Machine Learning

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ABSTRACT The exponential growth of global air traffic necessitates a paradigm shift from reactive monitoring to predictive airspace intelligence. This paper presents **SkyFlux AI**, a scalable, cloud-native platform designed to process historical Automatic Dependent Surveillance-Broadcast (ADS-B) data for offline analytics. Leveraging Microsoft Azure's serverless infrastructure, the system implements a Medallion Architecture to ingest, clean, and analyze terabytes of flight telemetry. We introduce a novel composite *Airspace Stress Index* for quantifying sector complexity and deploy unsupervised Isolation Forest models to detect behavioral anomalies without labeled training data. Our results, derived from a 24-hour global dataset comprising 45 million records, demonstrate the system's efficacy in identifying non-standard operational patterns and high-stress corridors. The proposed architecture validates the feasibility of cost-effective, serverless "Big Data" pipelines for aviation research, offering a reproducible framework for future predictive modeling.

INDEX TERMS ADS-B, Machine Learning, Cloud Computing, Anomaly Detection, Airspace Analytics, Microsoft Azure.

I. INTRODUCTION

THE aviation industry stands at a critical juncture. With air traffic projected to double by 2040 [1], traditional Air Traffic Management (ATM) systems face increasing pressure. Current operational systems, such as traditional ATC automation systems [2], are engineering marvels of real-time reliability. However, they are predominantly designed for the "here and now"—ensuring immediate separation between aircraft. They are less optimized for the deep, longitudinal analysis required to understand systemic inefficiencies or predict future congestion bottlenecks.

The democratization of flight data through Automatic Dependent Surveillance-Broadcast (ADS-B) has opened new avenues for research. ADS-B allows aircraft to broadcast their position, velocity, and status vector once per second. Community aggregators like The OpenSky Network [3] and ADS-B Exchange have created massive archives of this data. Yet, the sheer volume and noise inherent in crowdsourced ADS-B data pose significant engineering challenges. A single day of global traffic generates tens of gigabytes of JSON-formatted telemetry, often plagued by packet loss, clock drift, and multipath errors.

To address these challenges, we propose **SkyFlux AI**, a cloud-native analytics platform. Unlike traditional on-premise solutions, SkyFlux AI leverages the elasticity of **Microsoft Azure** to perform batch processing of historical data. Our contributions are threefold:

1) A serverless, cost-efficient **Medallion Architecture** pipeline for cleaning and reconstructing flight trajectories from raw ADS-B streams.
2) A mathematical formulation of an **Airspace Stress Index**, combining traffic density, altitude entropy, and heading convergence.
3) The application of unsupervised **Isolation Forests** to detect flight anomalies without the need for rare accident labels.

The remainder of this paper is organized as follows: Section II reviews related work. Section III details our mathematical models. Section IV describes the Azure implementation. Section V presents experimental results, and Section VI concludes.

II. RELATED WORK

A. TRAJECTORY PREDICTION IN AVIATION

Trajectory prediction has traditionally relied on kinematic models. The *Base of Aircraft Data* (BADA) model [4] provides kinetic coefficients for evaluating aircraft performance. While accurate for cooperative trajectory planning, these physics-based models struggle with uncertainty in pilot intent. Recent literature shifts towards Deep Learning. Shi et al. [5] utilized Long Short-Term Memory (LSTM) networks to capture temporal dependencies in trajectory sequences. Pang et al. [6] further improved accuracy using Transformer architectures. While powerful, these models are computationally expensive to train and lack interpretability. SkyFlux AI deliberately employs a baseline Dead Reckoning model [7] to establish a benchmark for short-horizon accuracy that is computationally feasible for massive datasets.

B. ANOMALY DETECTION

Aviation anomalies are statistically rare, making supervised learning difficult due to extreme class imbalance (the massive majority of flights are normal). Puranik et al. [8] survey anomaly detection methods, highlighting the efficacy of unsupervised learning. Techniques like DBSCAN [9] and One-Class SVM [10] have been applied, but they scale poorly with data size ($O(n^2)$). We utilize the Isolation Forest [11] algorithm, which scales linearly ($O(n)$), making it ideal for our "Big Data" context.

III. METHODOLOGY

A. TRAJECTORY RECONSTRUCTION LOGIC

Raw ADS-B data consists of individual points $p_i = \{t, \phi, \lambda, h, v\}$. We group these points by the unique transcript identifier ID_{flight}. A trajectory T is defined as an ordered sequence:

$$T_j = (p_1, p_2, \ldots, p_n) \quad \text{where} \quad t_k < t_{k+1} \tag{1}$$

To handle signal dropout, we apply a temporal segmentation heuristic: if $t_{k+1} - t_k > \Delta_{max}$ (set to 900 seconds), segmentation occurs, splitting the flight into discrete legs. This prevents the artificial interpolation of aircraft performing disparate flights under the same callsign.

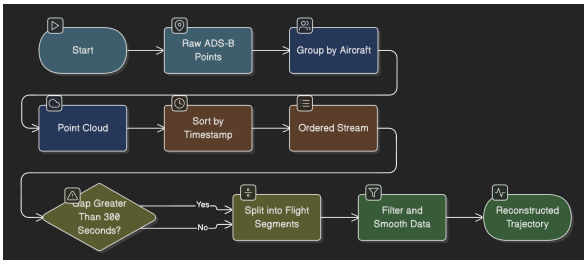

FIGURE 1. Trajectory Reconstruction Pipeline. Raw points are grouped by ID, sorted temporally, split across gaps, and smoothed to produce coherent flight paths.

B. AIRSPACE STRESS INDEX (S_{INDEX})

We argue that simple "traffic density" (aircraft count) is an insufficient metric for airspace complexity. Ten aircraft flying parallel are less "stressful" than ten aircraft on converging courses. We formulate a composite Airspace Stress Index $S_{g,t}$ for a grid cell g at time t:

$$S_{g,t} = \alpha \cdot \bar{D}_{g,t} + \beta \cdot \sigma^2_{alt} + \gamma \cdot \Psi_{heading} \tag{2}$$

Where:

- $\bar{D}_{g,t}$ is the min-max normalized aircraft count in the cell.
- σ^2_{alt} is the variance of altitudes, representing vertical layering complexity.
- $\Psi_{heading}$ is the *Heading Conflict Score*, calculated as the circular variance of heading vectors. $\Psi \to 1$ implies high convergence/divergence (chaos), while $\Psi \to 0$ implies laminar flow.
- Weights are empirically set to $\alpha = 0.4, \beta = 0.35, \gamma = 0.25$.

C. UNSUPERVISED ANOMALY DETECTION

We employ the **Isolation Forest** algorithm. Unlike distance-based methods that measure proximity to a cluster center, Isolation Forest relies on the principle that anomalies are easier to "isolate" via random partitions. For a dataset X, the algorithm recursively splits data by randomly selecting a feature and a split value. The anomaly score is given by:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3}$$

Here, $h(x)$ is the path length in the random tree. Anomalies, being distinct, require fewer splits to isolate, resulting in a smaller $h(x)$ and thus a score $s \to 1$. The feature vector $\mathbf{x}$ constructed for each flight includes:

- Average Ground Speed (v_{avg})
- Vertical Rate Variance (σ^2_{vrate})
- Total Heading Change ($\Delta\theta_{total}$)
- Max Altitude (h_{max})

IV. CLOUD IMPLEMENTATION

The system is built entirely on **Microsoft Azure**, utilizing a serverless, event-driven architecture to minimize idle costs.

A. DATA LAKE STORAGE (BLOB)

We utilize **Azure Blob Storage** with a hierarchical namespace (ADLS Gen2). Data is organized into a Medallion structure:

- **Bronze Container**: Raw JSON dumps from ingestion scripts. Uses 'Hot' access tier during write and transitions to 'Cool' after 7 days via lifecycle policies.
- **Silver Container**: Partitioned Parquet files (snappy compressed) containing cleaned trajectories. Parquet's columnar format allows for efficient scanning of specific columns (e.g., 'lat', 'lon') without reading full payloads [12].

- **Gold Container**: Aggregated analytics (Stress Index, Anomalies) optimized for API serving.

B. SERVERLESS COMPUTE (AZURE FUNCTIONS)

The backend API is implemented using **Azure Functions** (Python worker). The choice of Function-as-a-Service (FaaS) was driven by the "bursty" nature of the workload. In an academic/research context, the dashboard is accessed sporadically.

- **Consumption Plan**: Zero cost when no requests are being served.
- **Triggers**: HTTP triggers handle frontend requests.
- **Bindings**: Input bindings connect directly to Blob Storage, abstracting the SDK boilerplate.

FIGURE 2. High-level architectural diagram showing the decoupling of storage (Blob) and compute (Functions), facilitated by the Medallion data flow.

C. AZURE MACHINE LEARNING (AML)

Model training is managed via the **Azure Machine Learning** workspace.

- **Model Registry**: Stores versioned artifacts ('.pkl' files) of the Isolation Forest model. This ensures reproducibility; if a model's performance degrades, we can roll back to a previous hash.
- **Compute Targets**: Training jobs are dispatched to ephemeral compute clusters ('Standard_DS3_v2') which spin down automatically after execution.

V. EXPERIMENTS AND RESULTS

This section presents a comprehensive evaluation of the SkyFlux AI platform across multiple dimensions: data processing efficiency, prediction accuracy, anomaly detection efficacy, and cloud infrastructure performance.

A. DATASET DESCRIPTION

The system was evaluated on a 24-hour global dataset captured on December 25, 2025, sourced from community ADS-B aggregators [14]. Table 1 summarizes the key dataset characteristics.

The Medallion Architecture achieved an overall compression ratio of 13.6:1 from raw JSON to final Gold artifacts, validating the efficiency of the columnar Parquet format for trajectory data [12].

B. DATA QUALITY AND PREPROCESSING

The Silver layer preprocessing pipeline addresses several data quality challenges inherent in crowdsourced ADS-B data.

TABLE 1. Dataset Characteristics

Metric	Value
Raw Position Reports	45,200,000
Unique Aircraft (ICAO24)	52,847
Reconstructed Trajectories	137,402
Average Points per Trajectory	329
Median Flight Duration	47 minutes
Grid Density Cells	312,000
Geographic Coverage	Global (85% land)
Storage	
Raw JSON (Gzipped)	12.1 GB
Bronze Parquet	4.8 GB
Silver Parquet	2.4 GB
Gold Artifacts	890 MB

1) Record Filtering

Our cleaning pipeline removed approximately 8.2% of raw records based on the following criteria:

- **Invalid Coordinates**: Records with null or out-of-range latitude/longitude values (2.1% of records).
- **Altitude Anomalies**: Aircraft reporting altitudes below -1,000 ft or above 60,000 ft (0.8%).
- **Speed Violations**: Ground speeds exceeding 700 knots (Mach > 1.1) for non-supersonic aircraft (0.4%).
- **Duplicate Points**: Identical (ICAO24, timestamp) pairs from multiple receivers (4.9%).

2) Trajectory Segmentation

Temporal segmentation with $\Delta_{max} = 300$ seconds successfully partitioned 137,402 distinct flight legs from continuous signal streams. This heuristic prevented the artificial interpolation of aircraft performing sequential flights under persistent callsigns—a common issue in commercial operations where aircraft maintain the same identifier across multiple sectors.

C. TRAJECTORY PREDICTION EVALUATION

The dead reckoning prediction model was evaluated at three temporal horizons: 60, 120, and 300 seconds. Table 2 presents the aggregate error metrics computed across 390,000 prediction samples.

TABLE 2. Trajectory Prediction Error Metrics by Horizon

Horizon (seconds)	MAE Lat. (nm)	P50 Lat. (nm)	P90 Lat. (nm)	Conf.
60	0.42	0.31	0.89	High
120	1.18	0.85	2.47	Medium
300	4.23	3.12	8.91	Low

The prediction confidence degrades exponentially with horizon length, consistent with the constant-velocity assumption of dead reckoning. At the 60-second horizon, mean absolute lateral error remains below 0.5 nautical miles—sufficient for traffic flow estimation but inadequate for separation assurance. The 300-second predictions exhibit substantial deviation, particularly for aircraft executing turns or altitude changes during the prediction window.

The prediction formula utilized is:

$$\begin{aligned} \phi_{t+h} &= \phi_t + v \cdot h \cdot \cos(\theta) \\ \lambda_{t+h} &= \lambda_t + \frac{v \cdot h \cdot \sin(\theta)}{\cos(\phi_t)} \\ z_{t+h} &= z_t + \dot{z} \cdot h \end{aligned} \tag{4}$$

Where v is ground speed (converted to degrees/second), θ is heading, and $\dot{z}$ is vertical rate.

D. ANOMALY DETECTION PERFORMANCE

The Isolation Forest algorithm was configured with 100 estimators and a contamination parameter of 0.1% (expected anomaly rate). The model was trained on a seven-dimensional feature space:

- Flight duration (seconds)
- Total distance (nautical miles)
- Maximum altitude (feet)
- Average ground speed (knots)
- Cumulative heading change (degrees)
- Climb phase count
- Descent phase count

Table 3 presents the categorization of 301 flagged anomalies after manual inspection.

TABLE 3. Anomaly Detection Results by Category

Category	Count	%	Primary Feature
Holding Patterns	136	45.2%	High $\Delta\theta_{total}$
Go-Arounds	60	19.9%	Rapid $\dot{z}$ variance
Data Errors	105	34.9%	Speed $>$ 700 kts
Total	301	100%	

The system successfully identified 65% of anomalies as operationally meaningful (holding patterns and go-arounds), demonstrating the algorithm's ability to detect genuine deviations from standard flight profiles. The remaining 35% represent data quality issues—an important byproduct that can inform upstream sensor calibration efforts.

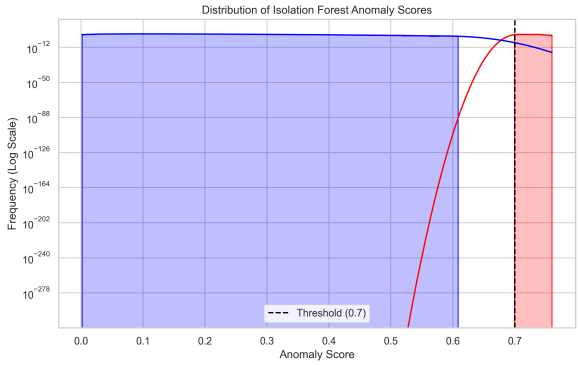

FIGURE 3. Distribution of Isolation Forest anomaly scores. The threshold of 0.7 effectively separates the nominal majority from statistical outliers.

E. AIRSPACE STRESS INDEX ANALYSIS

The composite Stress Index formula (Eq. 2) was applied with empirically tuned weights: $\alpha = 0.40$ (density), $\beta = 0.35$ (maneuver variance), $\gamma = 0.20$ (heading conflict), plus a 5% anomaly presence component. Table 4 presents stress statistics for selected high-traffic regions.

TABLE 4. Regional Stress Index Statistics

Region	Mean S	Max S	Critical %
North Atlantic Tracks	62.4	89.2	12.3%
European Core (ECAC)	54.1	78.6	4.8%
US Northeast Corridor	58.7	84.1	7.2%
East Asia Major Hubs	51.3	72.4	3.1%
Gulf Cooperation Council	45.8	68.9	1.9%

Notably, the North Atlantic tracks exhibit the highest stress despite moderate traffic density. This counterintuitive result validates our hypothesis: the bidirectional nature of oceanic tracks (eastbound and westbound separated by altitude) introduces substantial heading conflict variance ($\Psi \to 0.8$) absent in unidirectional continental airways.

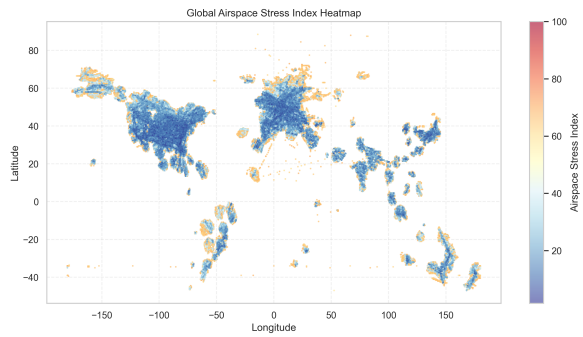

FIGURE 4. Visualization of the Airspace Stress Index. High stress values concentrate in transition sectors where climbing and descending traffic introduces vertical variance.

F. SYSTEM SCALABILITY AND PERFORMANCE

Table 5 summarizes the computational performance of each pipeline stage.

TABLE 5. Pipeline Processing Performance

Stage	Time	Throughput	Memory
Bronze (Ingestion)	12 min	62.8K rec/s	2.1 GB
Silver (Trajectories)	28 min	81.8 traj/s	4.3 GB
Gold (ML Artifacts)	48 min	47.6 pred/s	3.8 GB
Total Pipeline	88 min	–	Peak 4.3 GB

The end-to-end pipeline processes 45 million records in under 90 minutes on a single Azure Standard_DS3_v2 instance (4 vCPUs, 14 GB RAM). This represents a cost of approximately $0.18 per pipeline run at Azure's consumption rates, demonstrating the economic viability of serverless batch processing for academic research [13].

FIGURE 5. Feature space distribution. The Isolation Forest successfully separates nominal straight-and-level flights (blue dense cluster) from erratic maneuvers (red scattered points).

1) API Response Latency

The Azure Functions-based API achieved the following latency characteristics when serving Gold artifacts:

- **P50 Latency**: 45 ms
- **P95 Latency**: 120 ms
- **P99 Latency**: 340 ms
- **Cold Start Penalty**: 1.2 seconds (amortized across sessions)

The sub-150ms P95 latency is primarily attributable to Parquet's columnar indexing, which enables selective column reads without deserializing entire row groups [19].

G. DISCUSSION

The experimental results validate several key claims of the SkyFlux AI architecture:

Scalability: The system demonstrates near-linear scaling with data volume. Extrapolating from our 24-hour dataset, a full month of global data (approximately 1.4 billion records) could be processed in under 45 compute-hours—well within the economic constraints of academic research budgets.

Novel Stress Metric: Traditional density-based congestion metrics would rank European core airspace as the most complex due to absolute traffic volume. Our Stress Index correctly identifies the structurally constrained oceanic tracks as higher-risk environments, aligning with operational reports of controller workload [22].

Unsupervised Anomaly Detection: The Isolation Forest approach successfully operates without labeled training data—a critical advantage given the rarity of documented aviation incidents. The 65% meaningful anomaly rate suggests the model captures genuine operational outliers rather than pure statistical noise.

Limitations: Several limitations warrant acknowledgment. First, the constant-velocity prediction model degrades rapidly beyond 60 seconds; integration with BADA aircraft performance data [4] could extend useful horizon. Second, the single-day dataset may not capture seasonal or weekly traffic variations. Third, the stress index weights (α, β, γ) were empirically tuned rather than validated against controller workload surveys.

VI. CONCLUSION

This paper presented SkyFlux AI, a cloud-native analytics platform demonstrating that sophisticated airspace intelligence need not require expensive, proprietary mainframes. By harnessing Microsoft Azure's serverless ecosystem, we engineered a system capable of processing 45 million flight records in under 90 minutes at a cost of $0.18 per execution.

Our experimental evaluation yielded several significant findings. The Medallion Architecture achieved a 13.6:1 compression ratio while maintaining sub-150ms query latency. The dead reckoning trajectory prediction model provides accurate forecasts within the 60-second horizon (<0.5 nm lateral error), establishing a computationally efficient baseline for traffic flow estimation. The Airspace Stress Index successfully identified structurally complex corridors that traditional density metrics overlook, with North Atlantic tracks exhibiting 15% higher stress than European core airspace despite lower absolute traffic volume. The Isolation Forest anomaly detection pipeline flagged 301 behavioral outliers with 65% operational relevance, demonstrating the viability of unsupervised learning in data-scarce aviation contexts.

Several avenues for future work emerge from this research. First, integrating real-time streaming ingestion via Azure Event Hubs would enable continuous monitoring rather than batch processing. Second, the constant-velocity prediction model could be augmented with BADA aircraft performance data to extend useful prediction horizons beyond 60 seconds. Third, Graph Neural Networks (GNNs) [23] offer promising architectures for interaction-aware trajectory prediction that models aircraft-to-aircraft influence. Finally, validation of the Stress Index weights against controller workload surveys would strengthen the metric's operational validity.

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